Ultra-compact Reflow-compatible Deta Connector for Co-Packaged C

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Abstract: Ultra-compact detachable optical connector for Co-Packaged connector demonstrates low insertion loss ≤ 0.4 dB with small varia multiple cycles connections and 260 °C reflow-compatibility.

1. Introduction

Co-packaged Optics (CPO) is highly anticipated as a key enabler to realize higher consumption of optical communication in next generation data center (DC) [1, 2]. circuits (PICs) based on silicon photonic (SiPh) are placed close to application sp function as optical transceivers. For input and output of the light to and from PIC connectivity device for PIC/optical fiber connection is necessary. To avoid the co during the packaging process of CPO substrate, it is preferable to adopt a detachal remove the optical fiber side connector terminal during the packaging process. In is packaged by the reflow process, the reflow-compatibility is also required for th CPO substrate side. Furthermore, in recent years, it has been reported that a glass substrate is applied as a substrate for CPO [5]. The glass substrate has an advanta coefficient of thermal expansion (CTE). It is also possible to form an interface for connection with an external optical fiber by forming a waveguide inside of the gla

In this paper, we propose an ultra-compact and detachable optical connector f coupling by a collimated beam with magnetic force connection. We report on the reflow-compatibility. In addition, we demonstrate a concept of high-density multi connectors on glass substrate for CPO by taking advantage of the compact size of

2. Design concept of detachable optical connector employing collimated bean

In this paper, we develop an optical connector with a 12-channel, 250-μm pitch in concept for detachable waveguide/optical fiber connector. A waveguide in glass s connected with MT connector-like registration parts. The registration part is made and is manufactured by injection molding with high productivity. Lens arrays (LA the registration parts with reference to the guide mechanism and fixed, and the wa the FA are actively aligned and fixed to a rear surface of the LAs respectively. Th